

Operator Discovery Report: scaffold_selector_transfer_default

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Select a baseline TSP scaffold based on coarse instance structure descriptors to improve held-out transfer (bottlenecks/clusters/grids/traps/elongation).
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:cheapest_insertion_2opt.
- Surviving candidate: False.
- Problem class where it helps: heldout_tsplib, clustered_tsp, grid_like_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: b389b05ffb598dee.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Runtime inflation is high at 2.246598.

Transplant Results

- nearest_neighbor_2opt gap delta -0.024827, runtime delta 362.854686, distance-eval delta 255757.571429.
- cheapest_insertion_2opt gap delta 0.082494, runtime delta -3187.541014, distance-eval delta -6505300.428571.
- random_restart_2opt gap delta -0.004502, runtime delta 142.802614, distance-eval delta 239939.428571.
- sparse_three_opt gap delta -0.023542, runtime delta 358.383771, distance-eval delta 252336.285714.
- clustered_local_search gap delta -0.044792, runtime delta 148.947114, distance-eval delta 240221.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.210231 vs baseline 0.235058.

- clustered_tsp operator gap 0.038868 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.082744 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: -0.01625.
- Runtime inflation: 2.246598.
- Distance-evaluation inflation: 26.784097.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.016073.
- flattened_selector mean gap 0.153424 and delta vs full operator 0.04818.
- shuffled_selector mean gap 0.095997 and delta vs full operator -0.009247.